LoRa® Sensor Data with MKR WAN 1310

Learn how to send environmental data with LoRa® using two MKR WAN 1310 and a MKR ENV shield.

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In this tutorial, we will set up a configuration that allows two MKR WAN 1310's to send and receive sensor data. We will use the **LoRa** library to send data, and we will not use any external service. The sensor data will be recorded through the MKR ENV Shield, a shield that can record temperature, humidity, barometric pressure & ambient light.

Hardware & Software Needed

2x Arduino MKR WAN 1310 (link to store)

2x Antenna (link to store)

1x MKR ENV shield* (link to store)

2x Micro USB cable

Arduino IDE (offline and online versions available).

Arduino SAMD Board

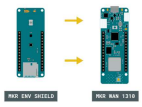
LoRa library installed, see the **github repository**.

Arduino_MKRENV installed, **click here for more documentation**.

> *This tutorial uses the MKR ENV shield which makes it easy to retrieve data from environmental sensors. If you have a temperature sensor or any other sensor you would like to use instead, you can make some easy adjustments to the code.*

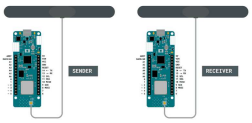
Circuit

First, if we are using the MKR ENV shield, simply attach it to one of the MKR WAN 1310 boards.



Attaching the shield.

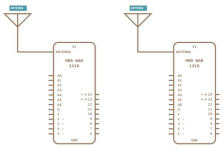
Follow the wiring diagram below to connect the antennas to the MKR WAN 1310 boards.



Simple circuit of the sender and receiver board with antenna.

Schematic

This is the schematic of our circuit.



Simple schematics of the circuit.

Let's Start

In this tutorial, we will use the MKR WAN 1310 board and the **LoRa** library to create a simple one way communication. It will teach you how to set up one board to be a **sender** and one board to be a **receiver**, and how to set up communication between these two without using any external services. We will learn how to send packages using the **LoRa** library, and how to receive packages.

All data received is printed in the Serial Monitor using a very simple interface.

To create the sender sketch, we will have to go through the following steps:

- Initialize the **SPI**, **LoRa** and **Arduino_MKRENV** libraries.
- Create a counter variable.
- Set the radio frequency to 868E6 (Europe) or 915E6 (North America).
- Read temperature from the MKR ENV shield.
- Begin a packet, print

Increase counter each loop and print it in Serial Monitor.

To create the receiver sketch, we will have to go through the following steps:

- Initialize the **SPI** and **LoRa** libraries.
- Set the radio frequency to 868E6 (Europe) or 915E6 (North America).
- Create a function to parse incoming packet.
- Print the incoming messages.

Code Explanation

We are going to program two separate MKR WAN 1310's in this tutorial. We will start with the **sender** device and later on, we will create a sketch for the **receiver** device.

Programming the Sender

The initialization is quick and easy: we will include the **SPI**, **LoRa** and **Arduino_MKRENV** libraries, and create a counter variable.

```
#include <Arduino_MKREN
#include <SPI.h>
#include <LoRa.h>

int counter = 0;
```

In the `setup()` we will begin serial communication, where we will use the command

program from running until we open the Serial Monitor.

We will then initialize the **LoRa** library, where we will set the radio frequency to 868E6, which is used in Europe for communication based on LoRa® technology. If we are located in North America, we need to change this to 915E6.

As we are using the MKR ENV shield, we also need to initialize the **Arduino_MKRENV** library by using the line `if (!ENV.begin())` followed by an error message in case it failed to initialize.

> *Note: If you are using another sensor, there is no need to include the Arduino_MKRENV library, or initialize it.*

```
void setup() {
  Serial.begin(9600);
  while (!Serial);
  Serial.println("LoRa

  if (!LoRa.begin(868E
    Serial.println("St
    while (1);
  }

delay(1000);

  if (!ENV.begin()) {
    Serial.println("Fa
    while (1);
  }
}
```

In the `loop()` we start by printing "Sending packet" in the

reading of the temperature sensor, using the command `double temperature = ENV.readTemperature();`. We use a double variable, as the temperature value we record has a decimal point (e.g. 25.85).

We then begin a packet by using the command, `LoRa.beginPacket()`, and print the temperature, followed by the message "celsius". This is done using the `LoRa.print()` function, which we then broadcast, using the `LoRa.endPacket()` command.

Finally, we add increase counter by 1 each time the loop has run, and a delay of 5 seconds, to limit the message rate. This will print in the Serial Monitor, to keep track on how many transmissions we have made.

```
void loop() {
  Serial.print("Sendin
  Serial.println(count

  double temperature =

  // send packet
  LoRa.beginPacket();
  LoRa.print(temperatu
  LoRa.print(" celsius
  LoRa.endPacket();
  counter++;
  delay(5000);
}
```

Programming the Receiver

The initialization and setup of the receiver is more or less identical to the **sender** sketch, but as we are writing this sketch to only receive data, we can remove the **Arduino_MKRENV** library.

Inside the loop, we will not be creating any packets. Instead, we will listen to incoming ones. This is done by first using the command `int packetSize = LoRa.parsePacket();`, and then check for an incoming packet. If we receive one, it is parsed, and printed in the Serial Monitor.

```
#include <SPI.h>
#include <LoRa.h>

void setup() {
  Serial.begin(9600);
  while (!Serial);

  Serial.println("LoR

  if (!LoRa.begin(868
    Serial.println("S
    while (1);
  }
}

void loop() {
  // try to parse pac
  int packetSize = Lo
  if (packetSize) {
    // received a pac
    Serial.print("Rec

    // read packet
    while (LoRa.avail
      Serial.print((c
    }

    // print RSSI of
```

Complete Code

If you choose to skip the code building section, the complete code can be found below:

Sender Code

MKR_WAN_13
Arduino MKR WA

MKR_WA Re

```
#include <Arduino_MKRENV.h>
#include <SPI.h>
#include <LoRa.h>

int counter = 0;

void setup() {

Serial.begin(9600);
  while (!Serial);

Serial.println("LoRa Sender");
```

Receiver Code

MKR_WAN_13
Arduino MKR WA

MKR_WA ☰ R

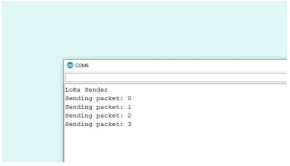
```
1   #include <SPI.h>
2   #include
    <LoRa.h>
3
4   void setup() {
5
    Serial.begin(960
    0);
6     while
    (!Serial);
7
8
    Serial.println("
    LoRa Receiver");
9
10    if
    (!LoRa.begin(868
```

Upload Sketch and Testing the Program

Once we are finished with the coding, we can upload the sketches to each board. The easiest way to go forward is to have two separate computers, as we will need to have the Serial Monitor open for both boards. Alternatively, we can use a Serial interfacing program called Putty. But for demonstration, it is good to use two computers. This way, you can move the boards further away from each other while testing the signal.

Sending Values

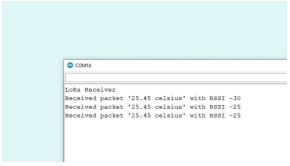
After we have uploaded the code to the **sender**, we need to open the Serial Monitor to initialize the program. If everything is working, it will start printing the message "Sending packet: x", where x represents the number of times the packet has been sent.



Sending packets.

Receiving Values

After we have uploaded the code to the **receiver**, we need to open the Serial Monitor to initialize the program. If everything works, we should now pick up the package we sent from the other device. The package contains the temperature, followed by RSSI (Received Signal Strength Indication). The closer this value is to 0, the stronger the signal are.



Receiving packets.

Experimenting with This Setup

Now that we have communication between the boards, we can do a

from the receiver device, we will start noticing changes in the RSSI. For example, while conducting this test, the sender device was moved around 20 meters away from the receiver, which decreased the RSSI to about -60.

Making Sender Device Mobile

As of now, we have a quite immobile setup, but this can be fixed by making the sender device mobile. A good way of testing this out is to make the sender device run on a battery instead. The MKR WAN 1310 has a battery connector that any battery with a JST PH connector can connect to. A standard 3.7 LiPo battery can power the MKR WAN 1310 for a longer time.

Troubleshoot

If the code is not working, there are some common issues we might need to troubleshoot:

- Antenna is not connected properly.
- The radio frequency is wrong. Remember, 868E6 for Europe and 915E6 for Australia & North America.
- We have not opened the Serial Monitor.
- We are using the same computer for both boards without a serial interfacing program.

Conclusion

This tutorial demonstrates a simple, yet powerful communication setup with the MKR WAN 1310 board and a MKR ENV Shield over the LoRa®-based network.

Trademark Acknowledgments

LoRa® is a registered trademark of Semtech Corporation.